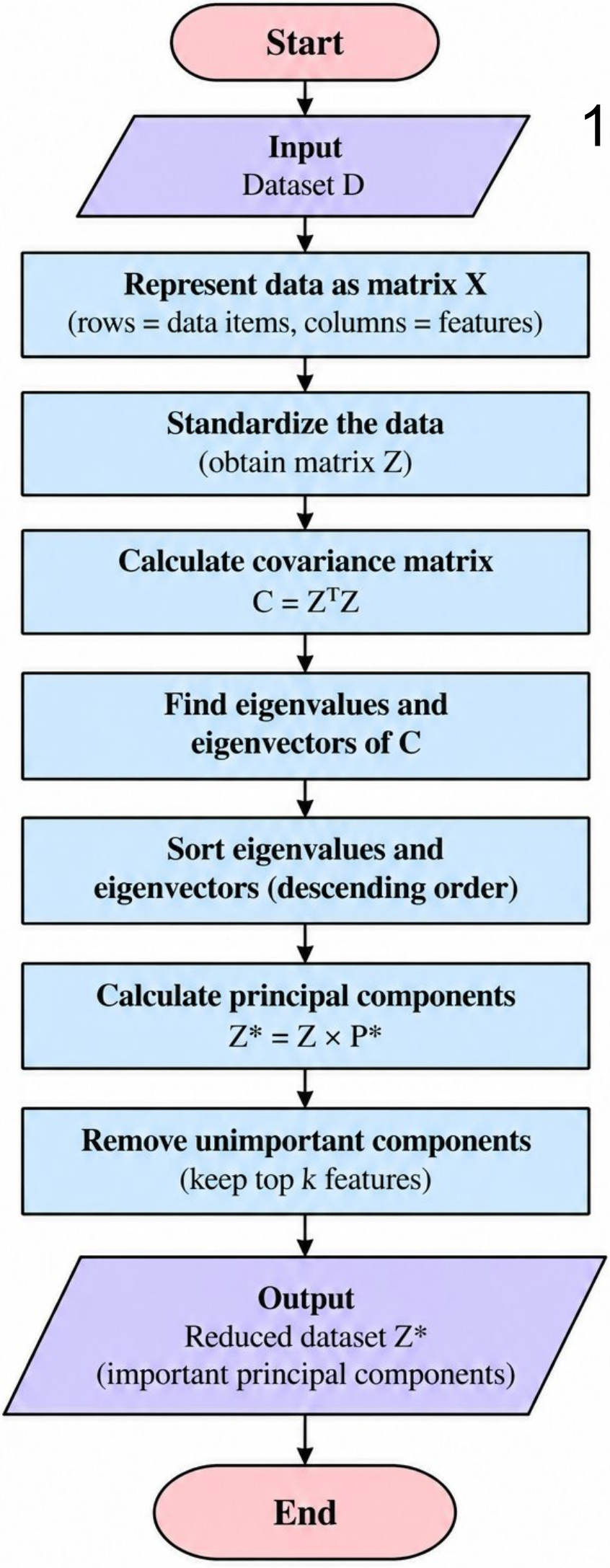
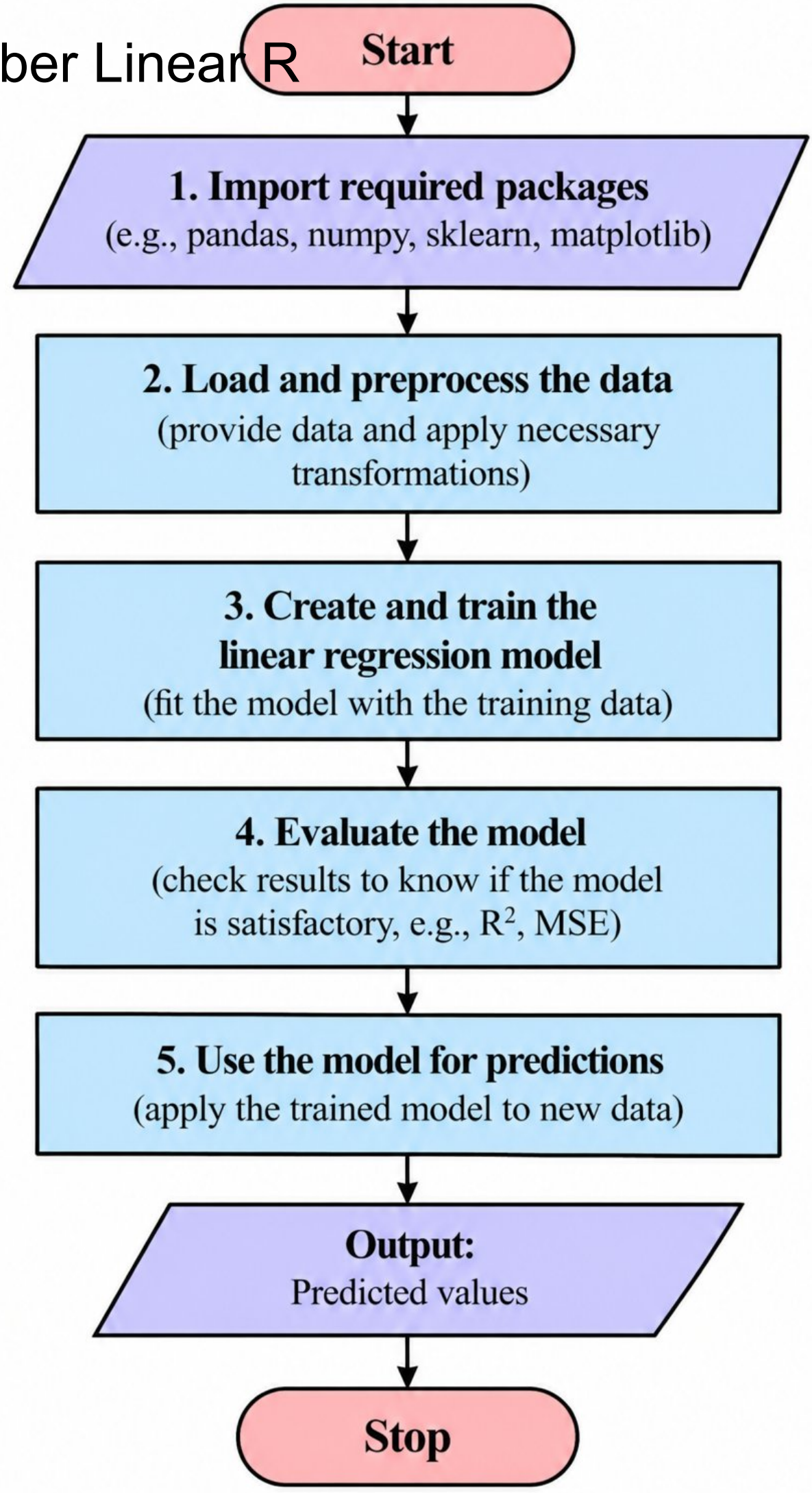
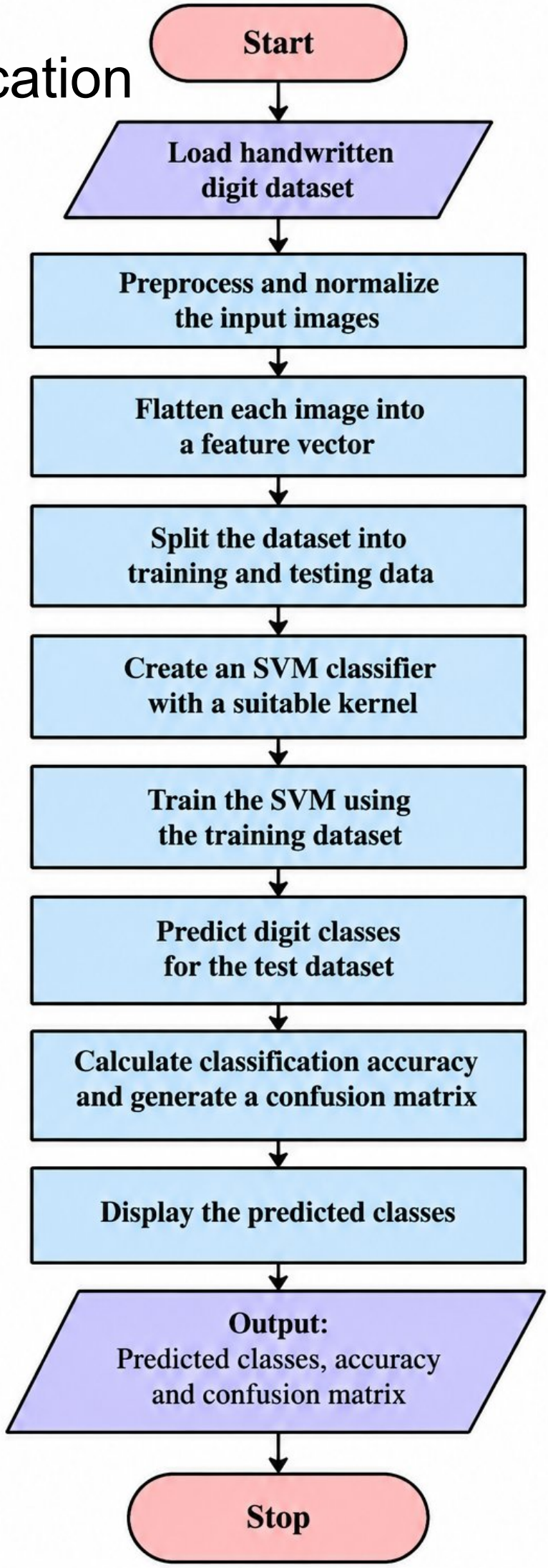
1) Wine


Start
Input
Dataset D
Represent data as matrix X
(rows = data items, columns = features)
Standardize the data
(obtain matrix Z)
Calculate covariance matrix
C = Z^T Z
Find eigenvalues and
eigenvectors of C
Sort eigenvalues and
eigenvectors (descending order)
Calculate principal components
Z* = Z × P*
Remove unimportant components
(keep top k features)
Output
Reduced dataset Z*
(important principal components)
End

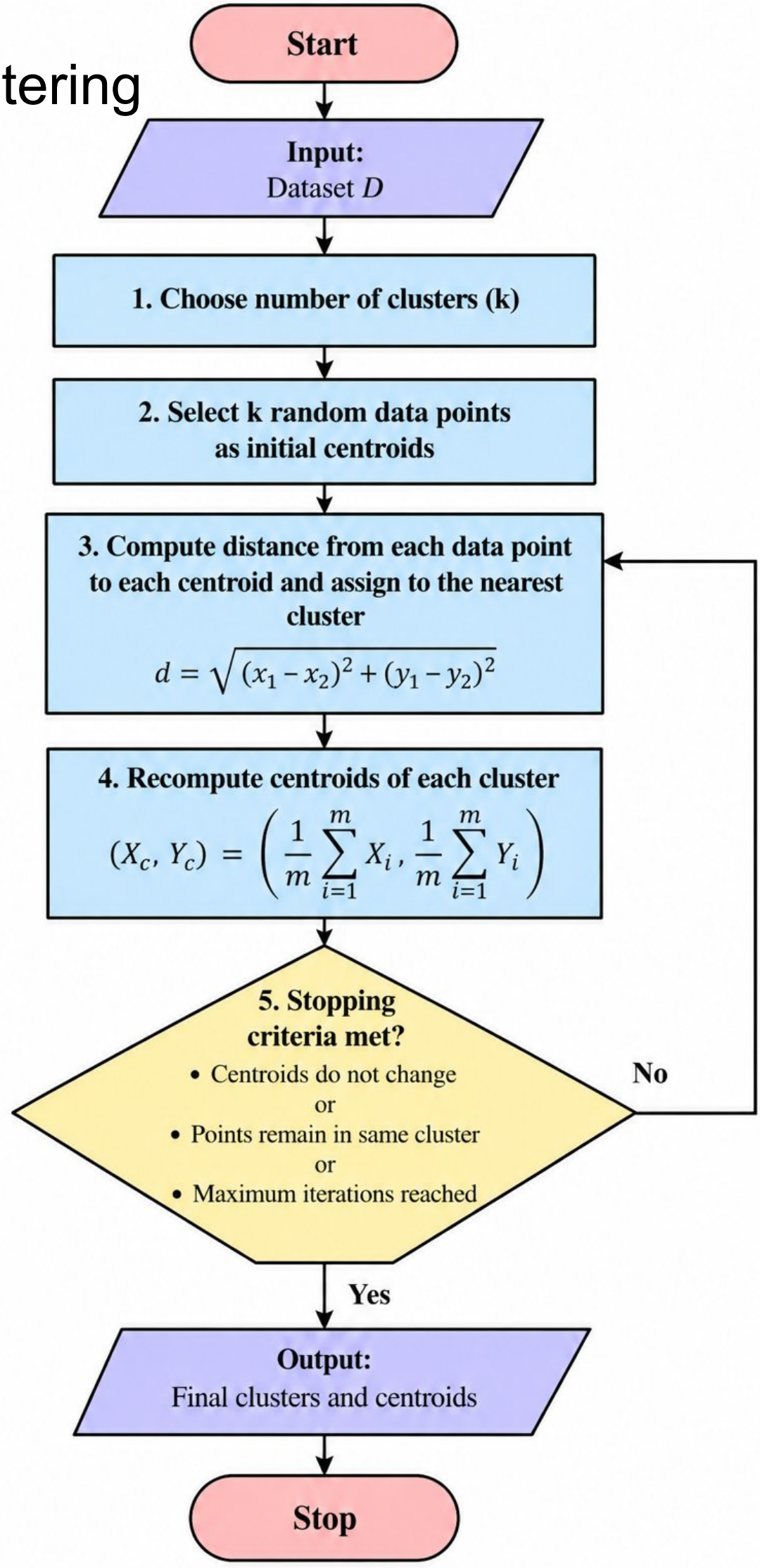
2)Uber Linear R



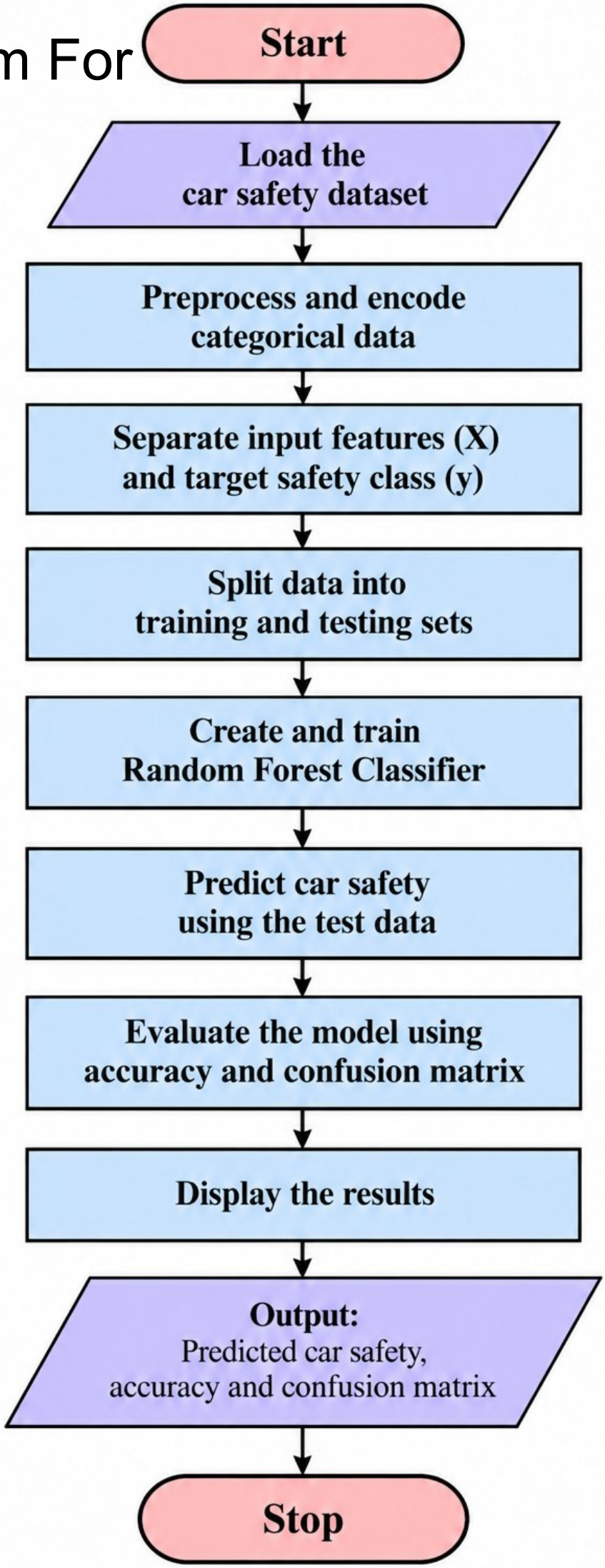
3)Classification



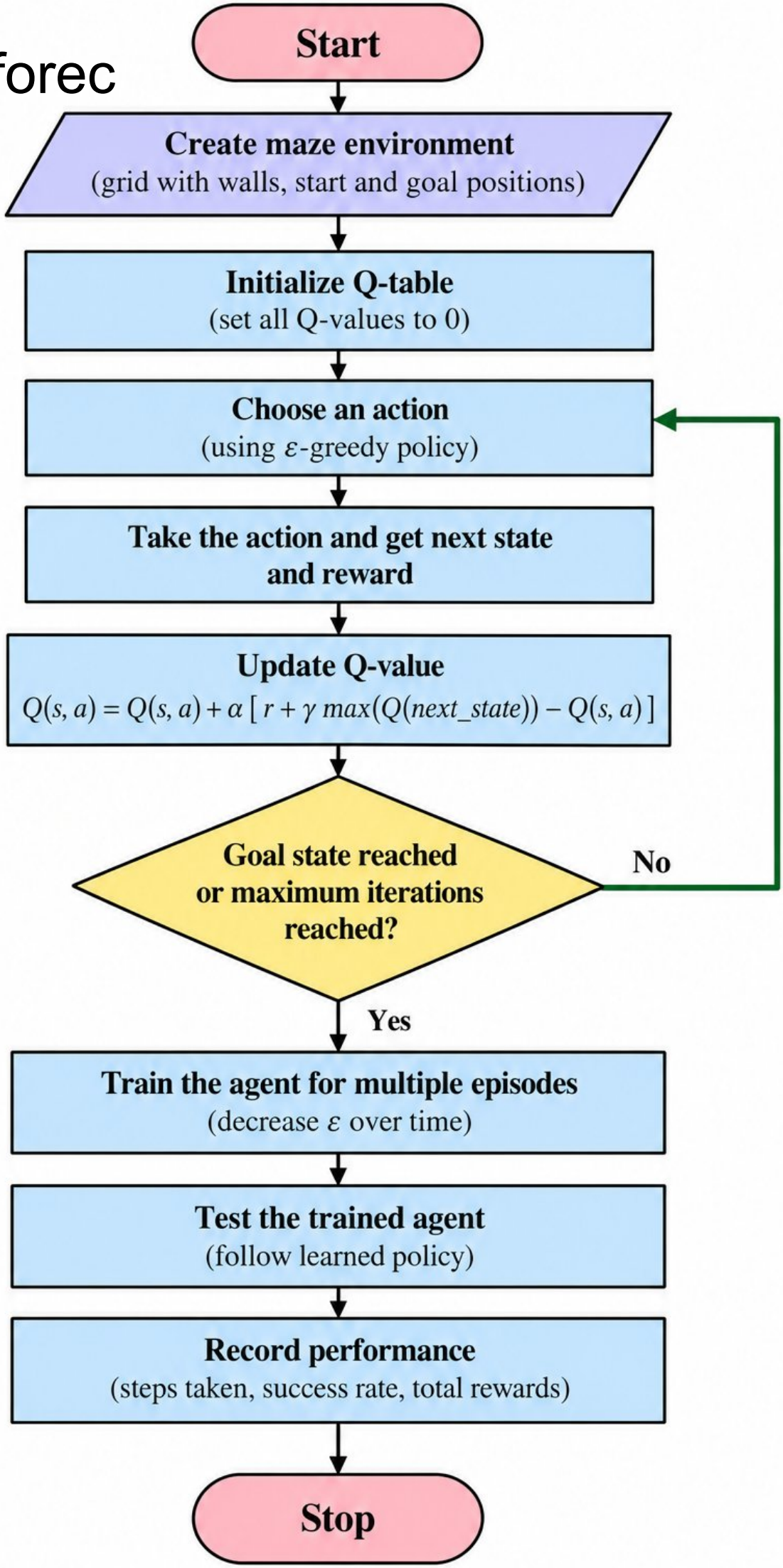
4)clustering



5)Random For


Start
Load the car safety dataset
Preprocess and encode categorical data
Separate input features (X) and target safety class (y)
Split data into training and testing sets
Create and train Random Forest Classifier
Predict car safety using the test data
Evaluate the model using accuracy and confusion matrix
Display the results
Output: Predicted car safety, accuracy and confusion matrix
Stop

6)Rainforec


Start
Create maze environment
(grid with walls, start and goal positions)
Initialize Q-table
(set all Q-values to 0)
Choose an action
(using ε-greedy policy)
Take the action and get next state
and reward
Update Q-value
$Q(s, a) = Q(s, a) + \alpha [r + \gamma \max(Q(next_state)) - Q(s, a)]$
Goal state reached
or maximum iterations
reached?
No
Yes
Train the agent for multiple episodes
(decrease ε over time)
Test the trained agent
(follow learned policy)
Record performance
(steps taken, success rate, total rewards)
Stop